

Muhammad Zawad Mahmud

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[LinkedIn](#) · [Google Scholar](#) · [GitHub](#) · [Website](#)

RESEARCH INTERESTS

Visual Place Recognition · Novel View Synthesis · Generative Models for Vision · Machine Learning Security

EDUCATION

M.S. in Computer Science, Fordham University, USA 2025 – May 2027 (expected)
CGPA 4.00 / 4.00

B.S. in Computer Science & Engineering, North South University, Bangladesh 2019 – 2023
CGPA 3.71 / 4.00 · Top 5% of class · Magna Cum Laude · Graduated with Academic Distinction

SELECTED PUBLICATIONS

1. *Hallucination-Aware Multi-View Fusion with Foundation Models: Robust Learning from Low-Quality Synthetic Views*. **M. Z. Mahmud**, D. Lyons, M. Rahouti, D. Camacho, A. Chehri. **Under review, Information Fusion**
2. *Understanding the Impact of Synthetic Novel-View Augmentation on Visual Place Recognition*. **M. Z. Mahmud**, S. Islam, D. Lyons, M. Rahouti, A. Chehri, K. Xiong. **Submitted, Pattern Recognition Letters**
3. *Membership Privacy Auditing for Machine Learning: Selection-Aware Multi-Attack Evaluation and DP-SGD Analysis*. M.Z.Mahmud, S. Islam, M. Rahouti, D. Verma, P. Novotny, A. Chehri. **Under review, ACM Transactions on AI Security and Privacy**
4. *Interpretable Dengue Detection Using XAI-Integrated Ensemble and Large Language Models on a Reduced-Feature Clinical Dataset*. **M. Z. Mahmud**, M. S. Reza, S. Islam, S. R. Alve, R. Khan. Array, vol. 31, art. 101120, 2026.
4. *Advanced Vision Transformers and Open-Set Learning for Robust Mosquito Classification*. A. A. J. Karim, **M. Z. Mahmud**, R. Khan. PLoS Computational Biology, 20(12), e1012654, 2024.

RESEARCH EXPERIENCE

Graduate Research Assistant, Dept. of Computer & Information Science, Fordham University Aug 2025 – Present

Advisors: Dr. Mohamed Rahouti, Dr. Damian Lyons

- Lead first-author work on generative view augmentation for visual place recognition (HAGVA). [A training-free gate scoring synthesized views by geometric consistency and semantic drift, recovering 78–118% of naive injection damage across three VPR datasets.]
- Developed a synthetic novel-view augmentation pipeline for VPR under viewpoint/appearance shift.
- Ongoing: auditing privacy leakage in federated learning via membership-inference attacks with differential-privacy defenses.

Research Assistant, North South University, Bangladesh

Jan 2024 – Aug 2025

Advisors: Dr. Riasat Khan, Dr. Md. Mamun Molla

- Produced applied-ML and explainable-AI work on clinical/biomedical classification, resulting in multiple peer-reviewed publications (see below).
- Mentored undergraduate researchers on Disease Classification.

TEACHING EXPERIENCE

- **Graduate Teaching Assistant**, Dept. of Computer & Information Science, Fordham University (Aug 2026 – Present).

- **Lab Instructor**, Dept. of ECE, North South University (Feb – Aug 2025).
- **Adjunct Lecturer**, Dept. of CSE, Green University of Bangladesh (Feb 2024 – Jun 2025); Structured Programming, Data Structures, DBMS, Data Communication.
- **Undergraduate Teaching Assistant**, North South University (Jan 2022 – Nov 2024).

ADDITIONAL PUBLICATIONS

First-author

Advance Transfer Learning Approach for Identification of Multiclass Skin Disease with LIME Explainable AI. **M. Z. Mahmud**, M. S. Reza, S. R. Alve, S. Islam, N. Fahmid. ICCIT 2024

Optimized IoT Intrusion Detection using Machine Learning. **M. Z. Mahmud**, S. Islam, S. R. Alve, A. J. Pial. IEEE RAAICON 2024

Predicting Adolescents' Attitudes Toward COVID-19 Vaccination Using Machine Learning. **M. Z. Mahmud**, I. Jahan, S. R. Alve, M. S. Reza, S. Islam. QPAIN 2025

Co-author

FairRate: A Practical and Fair Reputation Update Pipeline for Ride-Hailing Platforms. J. Bieniek, M. I. Mahmud, **M. Z. Mahmud**, S. Islam, D. C. Verma, M. Rahouti. **Under review, IEEE Intelligent Transportation Systems Magazine**

Enhancing Multi-Class Disease Classification Using Advanced LLMs. A. A. J. Karim, **M. Z. Mahmud**, S. Islam, A. Azam. ICCIT 2024

Deep Learning Approach for Oral Squamous Cell Carcinoma with LIME Explainable AI. S. Islam, **M. Z. Mahmud**, S. R. Alve, M. M. U. Chowdhury, F. I. Oishe. ICCIT 2024

Artificial Intelligence Driven Intrusion Detection for SDN Environments. S. Islam, S. R. Alve, **M. Z. Mahmud**, F. I. Oishe, M. M. Khan. ECCE 2025

Resource-Efficient Machine Learning for Multi-Class Threat Detection in IoT Environments. S. R. Alve, **M. Z. Mahmud**, S. Islam, Md. A. Chowdhury, J. Islam. QPAIN 2025

Interpretable Machine Learning for Osteoporosis Risk: Predictive Models and XAI Analysis. F. Elias, M. S. Reza, **M. Z. Mahmud**, S. Islam, S. R. Alve. QPAIN 2025

Optimized Malaria Identification through Transfer Learning. A. J. Pial, M. A. U. I. Sajid, A. Mubin, **M. Z. Mahmud**, M. F. Mridha, A. Jubair. ICCA 2024

TECHNICAL SKILLS

Languages: Python, C++, C

ML / DL: PyTorch, TensorFlow, Keras

Computer Vision / 3D: OpenCV, COLMAP, NeRF & Gaussian-Splatting frameworks, Open3D.

Tools / Infra: Linux, Git, Docker, HPC/SLURM.

AWARDS & STANDARDIZED TESTS

- Merit-Based Graduate Scholarship, Fordham University.
- 75% Merit Financial Aid, undergraduate (North South University).
- Magna Cum Laude (undergraduate CGPA ≥ 3.65).
- GRE 321 (V 159, Q 162, AWA 4.5) · TOEFL 107 (L 30, R 28, S 21, W 28).

ACADEMIC SERVICE

Reviewer: IEEE Transactions on Network and Service Management; Internet of Things & Cyber-Physical Systems (Elsevier); Next Material Journal (Elsevier); Array (Elsevier); Meteorological Applications (Wiley); IEEE Conference on Dependable and Secure Computing (DSC 2026); Intl. Conf. on Intelligent Data Analysis and Applications (IDAA 2025); International Conference on Electrical and Computer

Engineering Researches (ICECER 2026); International Conference on Computing Advancements (ICCA 2026).